

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2



ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H2 BIOLOGY

9648/2

Paper 2: Core

Tuesday

16 September 2011

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/11
3	/8
4	/13
5	/12
6	/13
7	/12
Total	/80

This document consists of **17** printed pages.

[Turn over

Section A

Answer all questions.

1. Fig. 1.1 below shows how the amount of DNA in a cell of *Chrysanthemum makinoi* varies during periods P to T in the cell cycle.

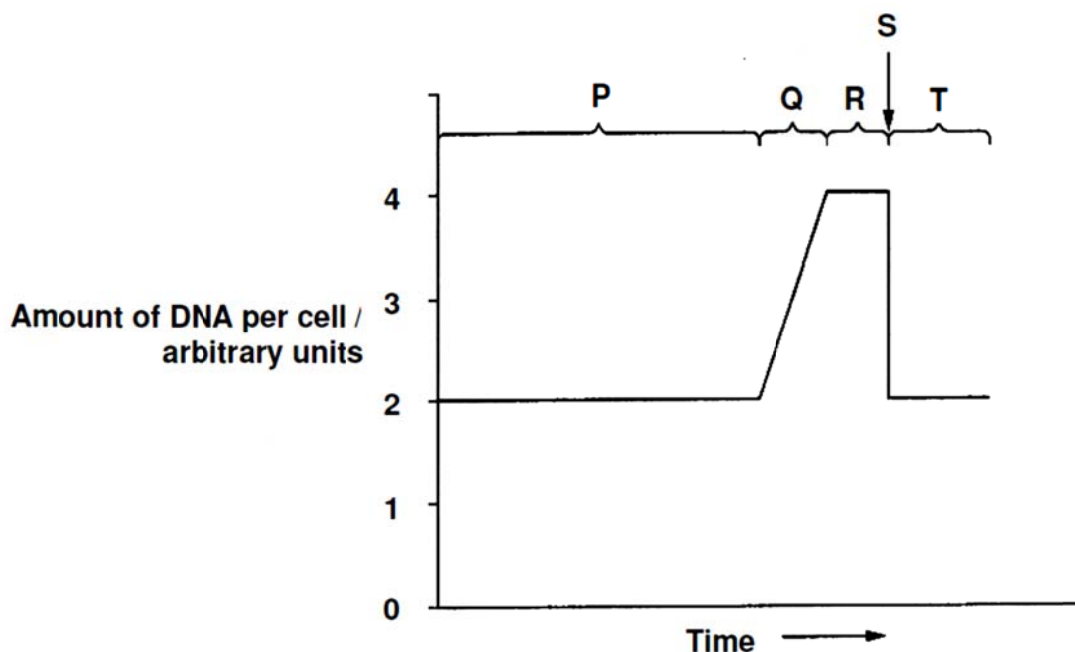


Figure 1.1

- (a) Radioactive thymine was supplied to the cells of some growing *Chrysanthemum* tissue. With reference to Fig. 1.1, state precisely during which period of the cell cycle does radioactivity of the nuclei first increase. Explain your answer.

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[2]

- (b) At anaphase of mitosis, sister chromatids separate to become individual chromosomes. Identify the period in Fig. 1.1 during which anaphase occurs. Briefly explain your answer.

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[3]

The first phase within interphase, from the end of the previous M phase until the beginning of DNA synthesis is called G_1 (G indicating *gap*). During this phase the biosynthetic activities of the cell, which had been considerably slowed down during M phase, resume at a high rate. This phase is marked by synthesis of various enzymes that are required in S phase, mainly those needed for DNA replication e.g. DNA polymerases known as Pol α .

(c) Describe briefly how the information on DNA is used to synthesize the RNA transcript for Pol α .

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(d) In eukaryotes, the RNA transcript synthesized after transcription is not suitable for immediate translation by the ribosomes. Describe the eukaryotic processing of pre-mRNA by snRNPs.

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[Q1 Total: 11]

2.

(a) Describe the structural features of the influenza genome.

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(b) Relate the structure of the influenza virus to its mechanism of penetration and release from the host.

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Pandemics are epidemics of infectious diseases that are spreading through human populations across a large region or even worldwide, usually costing many people to lose their lives e.g. the Spanish flu in 1918 caused the death of at least 50 million people worldwide. Throughout history, there had been a number of pandemics, such as smallpox or tuberculosis.

Fig. 2.1 highlights the significant dates in human influenza history.

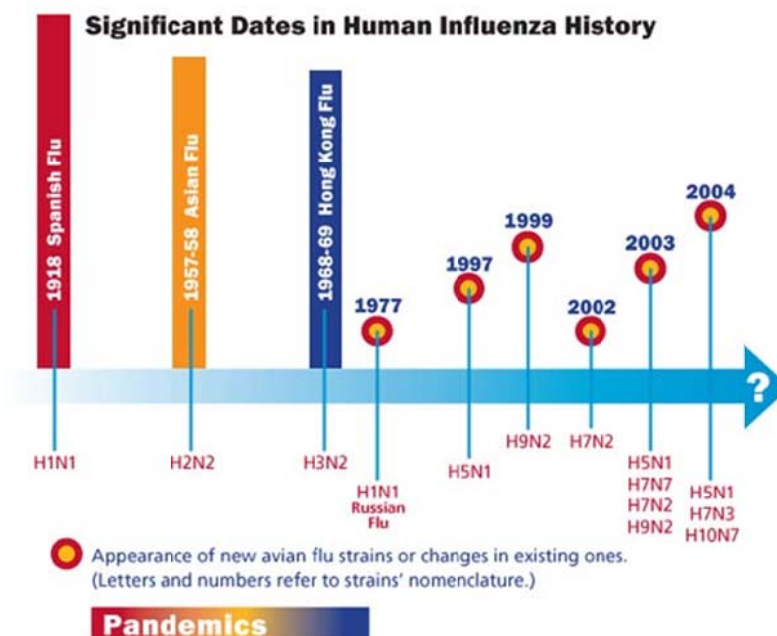


Figure 2.1

(c) Explain how the influenza pandemics arise.

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.....[4]

The Acquired Immune Deficiency Syndrome (AIDS) pandemic is a widespread disease caused by the Human Immunodeficiency Virus (HIV). Since AIDS was first recognized in 1981, it has led to the deaths of more than 25 million people, making it one.

(d) Other than the mechanism of penetration and release from the host cell, state one similarity and one difference between the reproductive cycle of the HIV and influenza viruses

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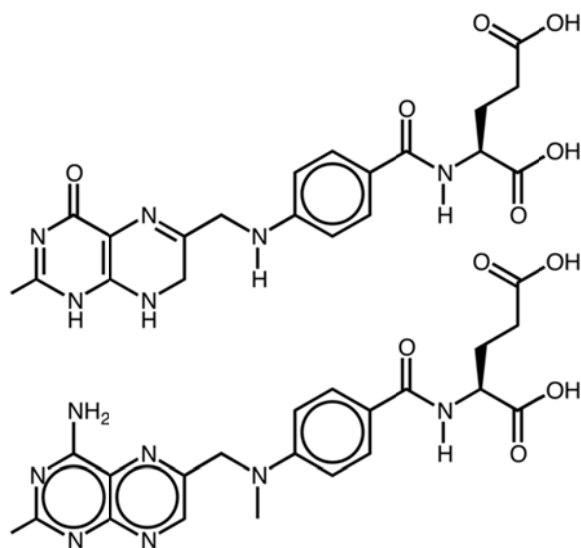
.....[2]

[Q2 Total: 11]

3.

(a) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This drug slows down cell division by acting on one of the enzymes dihydrofolate reductase, in a metabolic pathway. This enzyme catalyses the conversion of dihydrofolic acid to tetrahydrofolic acid, that results in the formation of purines. The chemical structures of both dihydrofolic acid and Methotrexate are shown in Fig. 3.1.



3.1
Structures of
dihydrofolic acid (top)
and Methotrexate
(bottom).

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

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(b) Binary fission differs significantly from the mitosis of eukaryotic cells. Describe the process of binary fission in bacterial cells.

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[Q3 Total: 8]

4. The chemical malvidin is one of the anthocyanidins responsible for the blue pigment found in the flowers of the *Primula* plants. Production of malvidin is controlled by two independent-assorting genes.

The dominant allele **R** codes for the production of malvidin. Plants with the dominant allele **D** results in the absence of malvidin production, even if they also carry the dominant allele **R**.

Non malvidin-producing Variety 1 has the genotype **RRDD** whilst non malvidin-producing plant Variety 2 has the genotype **rrdd**. The two varieties are crossed and the F_1 offspring intercrossed to give the F_2 generation.

- (a) (i) Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the F_1 and F_2 generations of the two crosses. Give the ratio of phenotypes in the F_2 generation. [6]

(ii) Suggest the molecular basis for this form of epistasis.

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(b) Pure breeding fruit flies, *Drosophila melanogaster*, with grey body and normal wings were crossed with pure breeding black body, vestigial wings. All the F₁ fruit flies had grey body and normal wings. Test cross of the F₁ fruit flies produced the following results:

grey body, normal wings	178
grey body, vestigial wings	132
black body, normal wings	142
black body, vestigial wings	160

(i) Calculate the value of χ^2 applicable to these data. (Workings need **not** be shown).

.....[1]

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

ν = degrees of freedom

E = expected 'value'

c = number of classes

A chi-squared table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(ii) Use the calculated value of χ^2 and the table of probabilities provided to find the probability of the results of the test cross departing significantly by chance from the expected ratio.

Probability =[1]

(ii) State what conclusions may be drawn from the probability found in **(b)(ii)**?

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[Q4 Total: 13]

5.

(a) An investigation into the production of ATP by mitochondria was carried out. ADP, phosphate, excess substrate and oxygen were added to a suspension of isolated mitochondria.

(i) Suggest the substrate used for this investigation.

.....[1]

(ii) Explain why the concentration of oxygen and amount of ADP fell during the investigation.

.....[2]

(iii) A further investigation was carried out to find out the effect of three inhibitors, **A**, **B** and **C**, on the electron transport chain in these mitochondria. In each of three experiments, a different inhibitor was added. The table below shows the state of the electron carriers, **W–Z**, after the addition of inhibitor.

Inhibitor added	Electron carrier			
	W	X	Y	Z
A	oxidised	reduced	reduced	oxidised
B	oxidised	oxidised	reduced	oxidised
C	reduced	reduced	reduced	oxidised

Give the order of the electron carriers in this electron transport chain.

Order [1]

(iv) Lactate dehydrogenase is responsible for the conversion of pyruvate to lactate by the enzyme. What would be the effect of inhibiting lactate dehydrogenase in a mammalian cell under anaerobic conditions?

.....[2]

(v) Plants can also respire anaerobically, but only for a short period of time. Explain why.

.....[1]

(b) Fig. 5.1 below shows the results of an experiment in photosynthesis using a unicellular green alga and the radioactive isotope ^{14}C to label carbon dioxide. The algae were actively photosynthesizing before the start of the experiment.

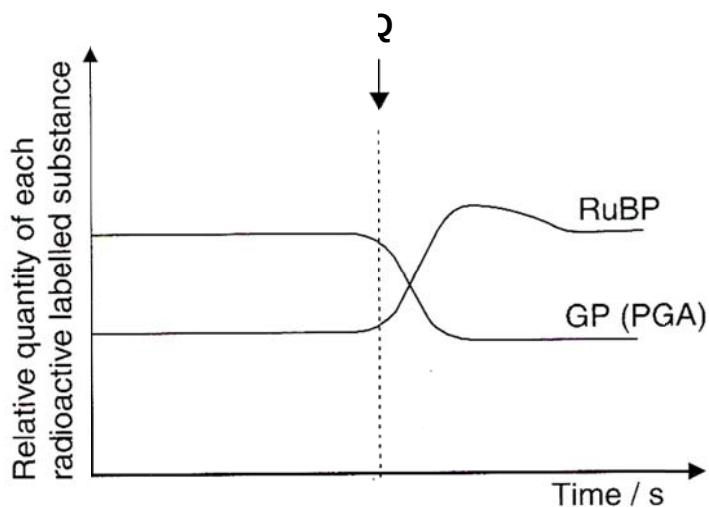


Figure 5.1

(i) State and explain what experimental condition was changed at Time Q.

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(ii) State two differences between the Krebs cycle and the Calvin cycle.

Krebs Cycle	Calvin Cycle

[2]

[Qn 5 - Total: 12]

Fig. 6.2 shows the generation of action potentials against time at the pre-synaptic terminal and a post-synaptic membrane.

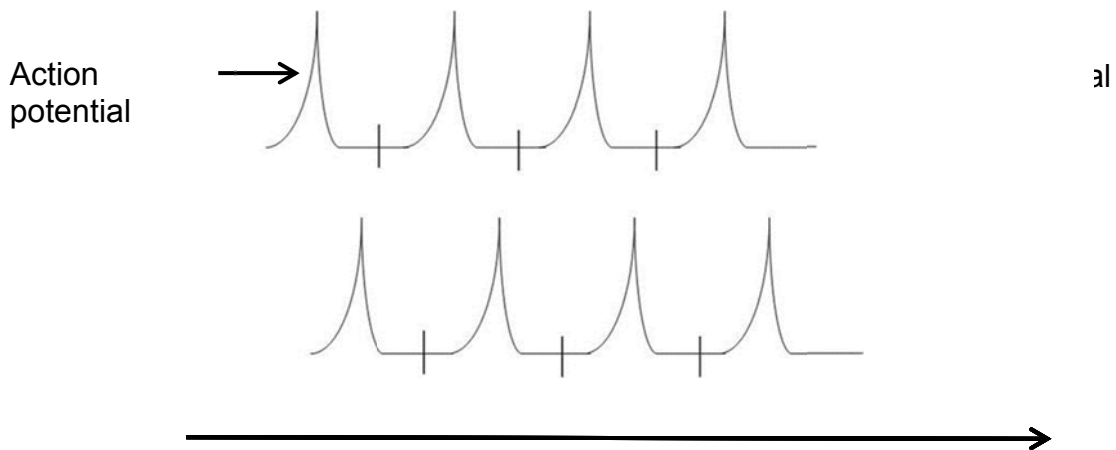


Figure 6.2

(c) With reference to the figure, account for the difference in the time of action potential generation.

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.....[2]

(d) A peptide has been developed in the pharmaceutical industry as a treatment for severe chronic pain. It is the synthetic equivalent of the venom of the marine snail (*Conus magus*), which induces paralysis in fishes. This peptide is found to affect voltage-sensitive calcium channels of neurones.

Describe how this peptide can result in pain relief.

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[Q6 Total: 13]

7. Lake Tanganyika is the deepest lake in Africa and the second largest lake by volume in the world. Scientists estimate the age of the lake to be between 9-12 million years old. Lake Tanganyika contains 3 major basins which formed separately and joined when their water rose over time.

The lake holds at least 250 species of cichlid fish and 150 non-cichlid species, which are highly diverse in terms of morphology, behaviour, and ecology. Almost all (98%) of the Tanganyikan cichlid species are endemic¹ to the lake.

Among the many species of cichlids, seven species in the genus *Perissodus* are known for their unique feeding habit of scale eating with specialized feeding morphology and behaviour.

Fig 7.1 shows the phylogenetic tree constructed based on the feeding habits (A) and the habitat depth and oral tooth shape (B) of nine species of *Perissodini*.

¹ Endemic- the ecological state of being unique to a defined geographic location.

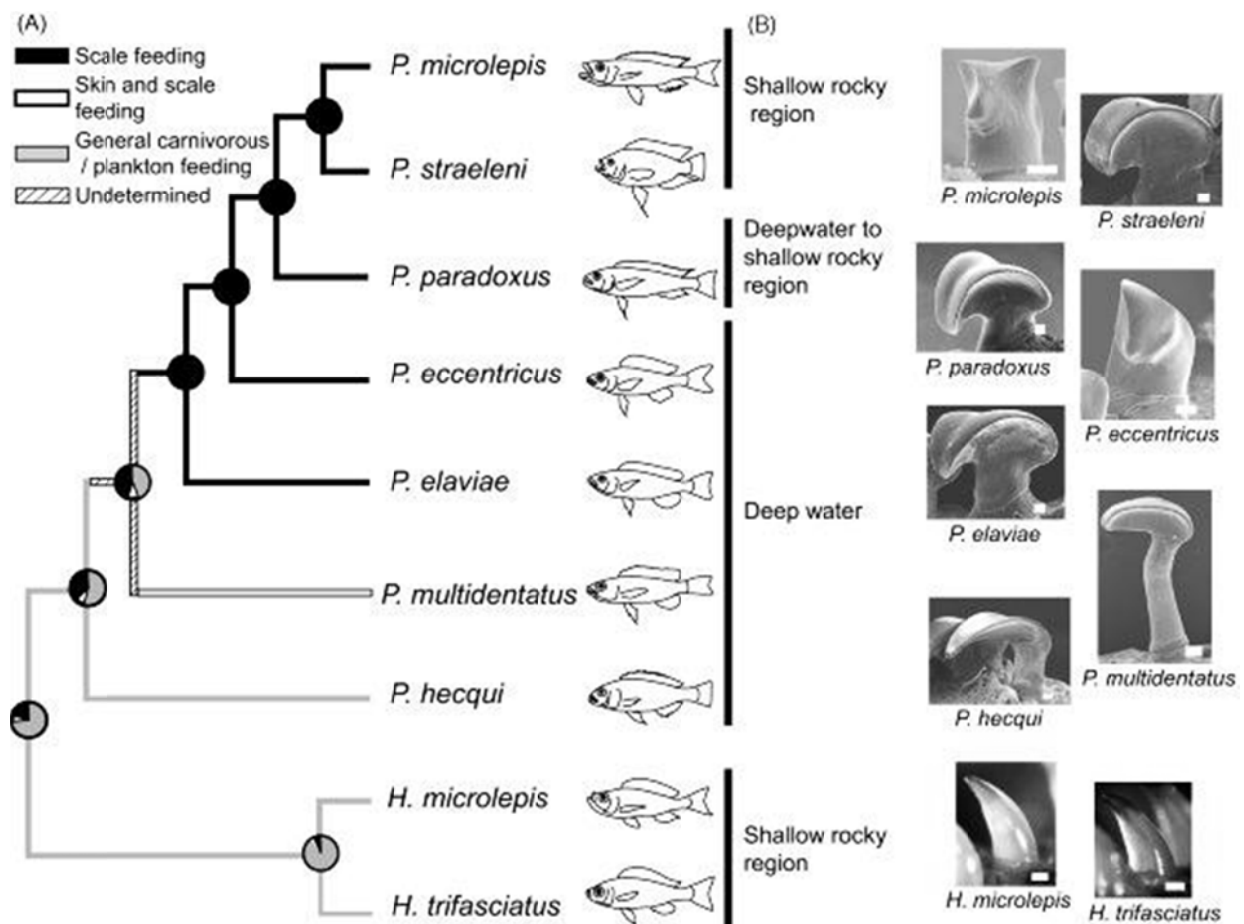


Figure 7.1

(a) Explain how natural selection could have caused the diversity of oral tooth shape in the different cichlid species in Lake Tanganyika.

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(b) Suggest a reason why it may be more reliable to construct a phylogenetic tree of the nine species of *Perissodini* using molecular data.

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(c) Based on the fossil record, *P. microlepis* and *P. straeleni* share a common ancestor about 50,000 years ago. Describe how the molecular clock can be used to verify this information.

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A most peculiar behavior is seen in the scale-eating cichlid, *P. microlepis*, which feeds exclusively on the scales of other cichlids to survive. Phenotypes of this cichlid species are either right-mouthed or left-mouthed.

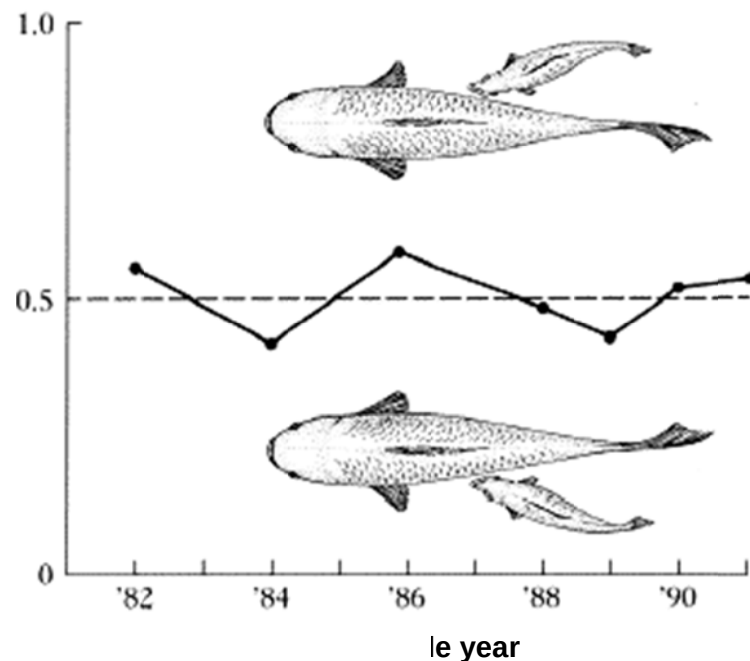


Figure 7.2

(d) The relative frequency of the left-mouthed fish and right-mouthed fish fluctuate around a mean of 0.5 (see Fig 7.2) and the presence of both phenotypes appears to be in an evolutionary stable state. Suggest an explanation for this phenomenon.

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[3]

(e) After a major drought, there was a sharp increase in the number of left-mouthed fish which lasted for a long period of time. Explain why.

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[2]

[Q7 Total: 12]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell. [8]
- (b)** Contrast triglycerides and phospholipids. [4]
- (c)** Describe the main features of collagen that contribute to its tensile strength [8]

[Total: 20]

- 9** **(a)** Compare the structure and organisation of prokaryotic and eukaryotic chromosome. [6]
- (b)** Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [8]
- (c)** Describe the development of cancer as a multi-step process. [6]

[Total: 20]